

# Sea Terp

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FIRE

## Background

- ❖ **Dead Zones** are areas of water with little to no oxygen. Dead zones typically form in deeper waters and imbalance marine ecosystems<sup>5</sup>
- ❖ Habitats in the **Chesapeake Bay** are susceptible to pollutants and low oxygen levels
  - The **largest dead zone** in the Chesapeake Bay spans **1.2 cubic miles**

## Inspiration

We drew inspiration from the **green sea turtle** for their efficient underwater mobility and hydrodynamic shells



- ❖ Sea turtles can **alter** their **buoyancy** in the water allowing them to dive and rise with ease<sup>3</sup>
- ❖ Sea turtles use their **long front flippers** to accelerate

## Project Goals

- ❖ Build a robot with **sensors** able to **assess water health** and diagnose issues
- ❖ The robot must endure a wide range of **temperatures** and variable **water pressure**
- ❖ The robot will also interact with **marine life** like fish and crabs.
- ❖ Mimic sea turtle's **swimming mechanisms** and hydrodynamic body shape

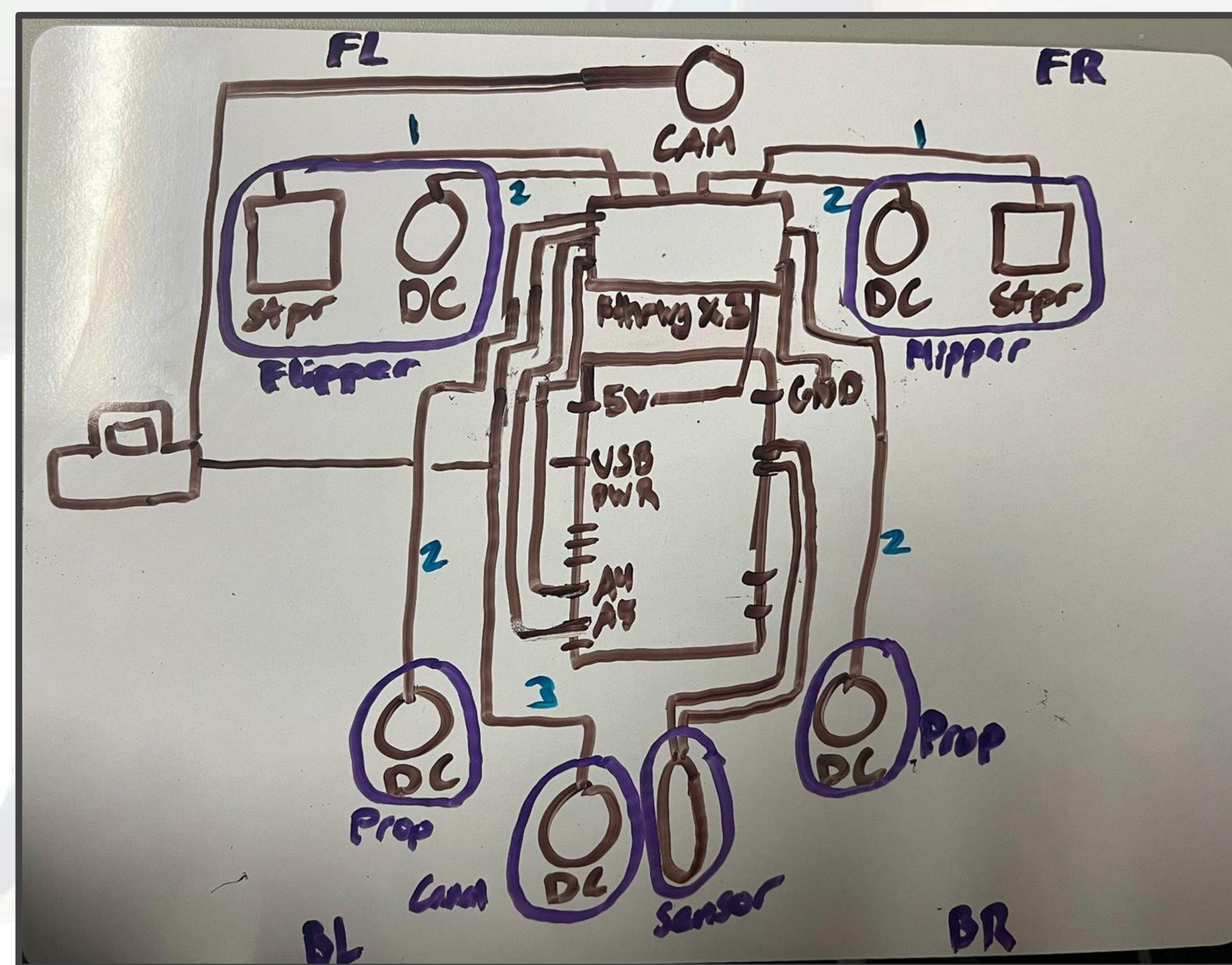
## Project Design

Sea Terp features the following:

- ❖ Arduino-compatible **oxygen sensor** to monitor water quality coupled with an interior **canal** for water flow
- ❖ **Camera** to collect visual **data samples** from dead zones and other habitats
- ❖ **Buoyancy chamber** to ascend and descend underwater
- ❖ **Bio-inspired flippers** for steering
- ❖ **Bio-inspired shell** to house sensors and provide hydrodynamic shape
- ❖ Rear-mounted **propellers** to provide thrust

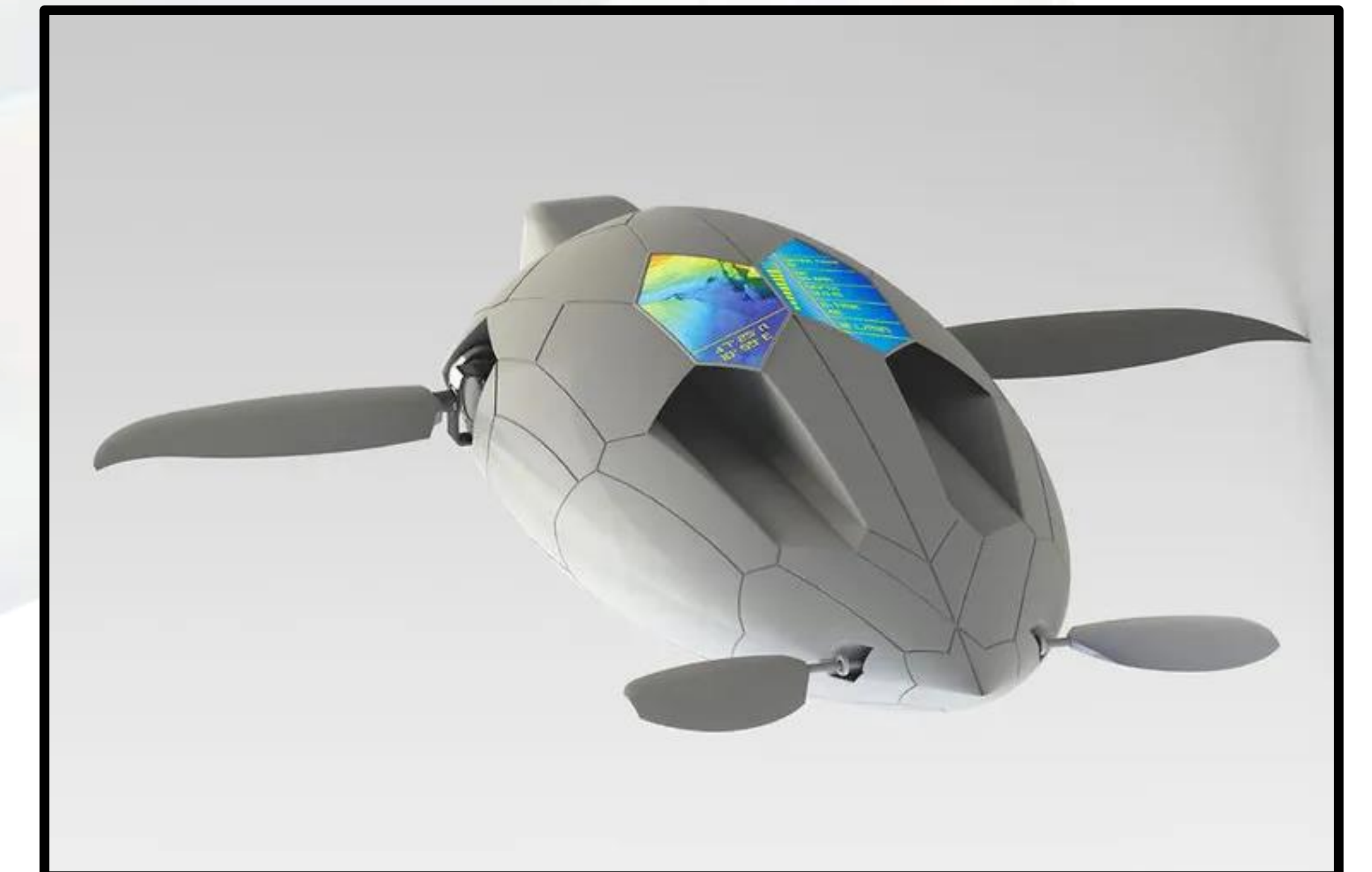


## Electronics



## Future Steps

- ❖ **Prototype** Sea Terp's swimming mechanism, sensor suite, communication system, and power source
- ❖ Test each component individually to ensure **functionality and reliability**
- ❖ Run initial tests of prototype in **controlled laboratory environments** to assess its abilities
- ❖ Conduct **field trials** of the robot in the **Chesapeake Bay** to validate its performance in real-world conditions
- ❖ Program a **remote controller** to allow for Sea Terp to be manually controlled.
- ❖ Design a **crankshaft system** to allow for a **greater range of motion** for the flippers.



## References

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